

# Horizontal IoT Platforms: An Overview

The Internet of Things (IoT) along with Big Data is captivating the digital world. A horizontal IoT platform encourages a large number of clients to work with a typical framework, thus enabling rapid growth and innovation in businesses. By making the gateway and cloud resources have open functionality, innovators can concentrate their efforts on creating devices and services.



**T**he backbone of IoT is connectivity; hence, an IoT platform can be as simple as software infrastructure that handles connectivity between physical objects. Due to a rise in the number of connected endpoints and smart devices, the IoT market is placing a huge demand on platforms that provide real-time intelligence, machine-to-machine interaction, cross-industry applications, multi-vendor interoperability and end-to-end security. Hence, the platform may be furnished with entities such as artificial intelligence (AI), machine learning and augmented reality. IoT basically consists of the following building blocks.

- **Connectivity:** All IoT platforms

have a connectivity layer that provides a duct for data acquisition, remote access and control.

- **Device management:** Provisioning tasks, device registry, device identification, device modelling, remote administration, software updating, authentication of cloud/backends, centralised log collection and management, fault tolerance, troubleshooting, diagnostics and remote reboot, are all tasks that are handled by the device management module.
- **Data management:** The data management block refers to the expansion, execution and administration of architectures,

software and disciplines that control, defend, deliver and improve the value of data and information assets.

- **Application enablement:** Application enablement provides action management to expound rules, and activates the automation of tasks and services.
- **Cloud back-end:** Cloud back-end has the ability to handle extensive amounts of data and is cost-effective in comparison with on-premise infrastructures. Cloud computing is deemed an essential part of commercial IoT.
- **Security:** We need to provide lots of security, trust, privacy and identity management across endpoint devices, the network and the cloud in IoT platforms. This is performed by the Security Block.

## Types of IoT platforms

IoT platforms can be categorised into different genres based on the unique types of services provided to end users. Platforms for smart wearables and home automation aim at the consumer market, and on unidirectional, data-to-cloud connectivity. On the other hand, Industrial IoT (IIoT) customers from sectors such as military, manufacturing, energy, automotive, agriculture, aerospace and banking are concerned with embedded software, tailored applications, device management, data management, interoperability, security, reliability

